

Multi-View Coverage Construction of MLFF Training-Set Size Ladders in mdstats

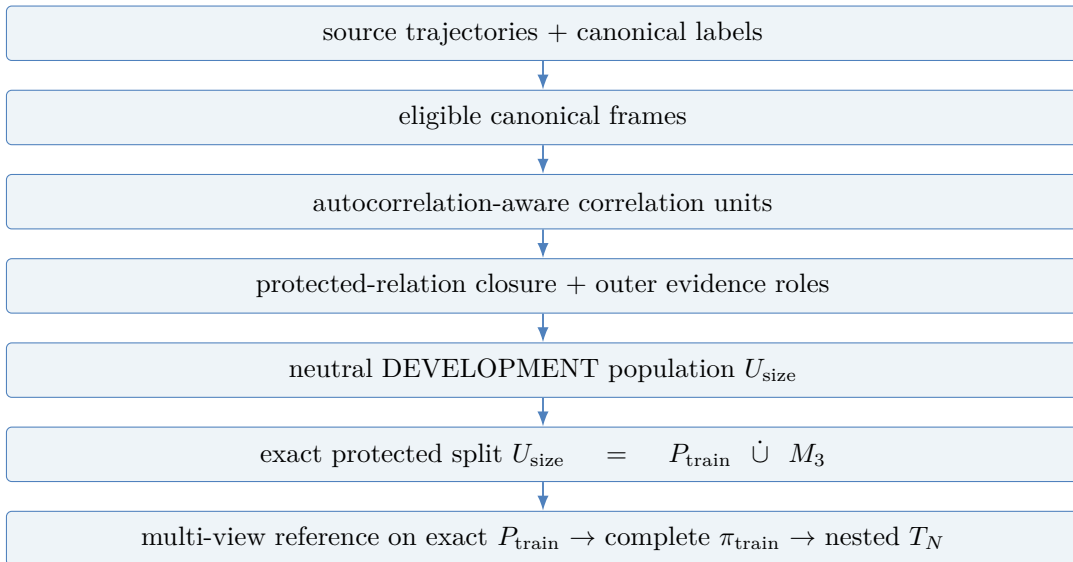
A concise method paper for a physics audience

From correlated MD trajectories to leakage-controlled, coverage-balanced nested training sets

Central idea. The `mdstats` MLFF workflow does not treat stored molecular-dynamics (MD) frames as independent observations and does not choose training configurations by uniform subsampling. It first converts eligible trajectories into autocorrelation-aware **correlation units**, protects additional relations that would make a train/evaluation split leak information, constructs the exact target-training population, and only then builds one deterministic multi-view ordering whose prefixes progressively cover the physically relevant variation of that population.

End-to-end preparation of the target-training population

The target-size experiment is downstream of several preparation steps. Evaluation, calibration, locked-test, and later cross-validation evidence are separated before target membership is constructed, so they cannot leak backward into the training-set ladder.



Conceptual preparation chain. The target-order selector begins only after the leakage-controlled P_{train}/M_3 split is frozen.

Let

$$U_{\text{size}} = P_{\text{train}} \dot{\cup} M_3$$

be the exact target-size development split. Here, M_3 is the largest P3 model-selection reserve and P_{train} is the sole domain from which target-training subsets are constructed. The ladder is therefore not a resampling of the entire trajectory archive.

Why MD frames must first be grouped statistically

Adjacent MD frames are serially correlated. If a trajectory is saved much more frequently than its slow physical degrees of freedom decorrelate, then many stored configurations describe essentially the same

local state. Treating those frames as independent would cause densely sampled trajectory segments to dominate both data splitting and feature-space coverage.

For a stationary scalar observable X_t ,

$$\rho(k) = \frac{\text{Cov}(X_t, X_{t+k})}{\text{Var}(X_t)}, \quad \tau_{\text{int}} = \frac{1}{2} + \sum_{k=1}^{k^*} \rho(k),$$

and

$$N_{\text{eff}} = \min\left(N, \frac{N}{2\tau_{\text{int}}}\right).$$

For an uncorrelated sequence under this convention, $\tau_{\text{int}} = 1/2$ and $N_{\text{eff}} = N$. A larger τ_{int} indicates increasing temporal redundancy. This is a *diagnostic* of serial correlation in the chosen observables, not proof that all slow structural variables have become independent.

How mdstats estimates the correlation time

For a finite sequence $x_0, \dots, x_{N-1}$ with mean $\bar{x}$,

$$\hat{\gamma}(k) = \frac{1}{N-k} \sum_{t=0}^{N-k-1} (x_t - \bar{x})(x_{t+k} - \bar{x}), \quad \hat{\rho}(k) = \frac{\hat{\gamma}(k)}{\hat{\gamma}(0)}.$$

The noisy long-lag tail is truncated using Geyer’s initial-positive-sequence rule: adjacent lag pairs are retained while

$$\hat{\rho}(2m-1) + \hat{\rho}(2m) > 0.$$

If K_+ is the retained lag set,

$$\hat{\tau}_{\text{int}} = \max\left(\frac{1}{2}, \frac{1}{2} + \sum_{k \in K_+} \hat{\rho}(k)\right).$$

No autocorrelation is computed across a source gap, continuation reset, or excluded interval.

From correlation time to a complete-frame block length

For each source run and its contiguous eligible segments, let j index configured observables and r index contiguous segments. The conservative accepted time is

$$\tau_{\text{max}} = \max_{j,r} \hat{\tau}_{j,r}.$$

With correlation multiplier $m_{\text{corr}} > 0$ and minimum block length L_{min} ,

$$L_{\text{corr}} = \max(1, \lceil m_{\text{corr}} \tau_{\text{max}} \rceil), \quad L = \max(L_{\text{min}}, L_{\text{corr}}).$$

All lengths are in **stored-frame units**. If frames are written every Δt_{save} , a block of L frames corresponds approximately to $L\Delta t_{\text{save}}$ of trajectory time.

For a contiguous eligible run of $n > L$ frames,

$$B = \max\left(1, \left\lfloor \frac{n}{L} \right\rfloor\right)$$

blocks are formed. Writing $n = qB + r$, the first r blocks contain $q + 1$ frames and the remaining blocks contain q . Every eligible frame is retained exactly once; no tail is discarded.

Current default policy binding. The current neutral implementation uses $m_{\text{corr}} = 2$ and $L_{\text{min}} = 32$ stored frames. Its default autocorrelation channels are energy per atom, force-component RMS, pressure, instantaneous temperature, and cell volume when available for the run; if none is complete, cell volume is used as the fallback observable. These are current configurable policy/default bindings, not universal physical constants.

What exactly is a correlation unit?

The block plan is only the first step. The neutral preparation layer enforces the physical condition identity

$$\kappa(x) = (\text{composition, temperature condition, strain class, regime, user labels}).$$

A final unit cannot cross a condition boundary, so candidate blocks are split where this identity changes. Conversely, if a recognized protected event spans neighboring blocks, those blocks are merged so the whole event remains indivisible. A protected event that crosses a declared condition boundary is inconsistent input rather than something silently split.

The resulting **correlation unit** is an indivisible, contiguous, complete-frame interval tied to one source run and one condition, constructed from autocorrelation-aware block evidence and enlarged when necessary to preserve a protected event. Each eligible frame belongs to exactly one such unit.

A correlation unit is *not automatically claimed to be an independent physical realization*. **mdstats** separately records the strongest available independence evidence: independent replicas, independent structural realizations, independent thermodynamic runs, purged temporal blocks, slow-state-not-decorrelated evidence, or insufficient-independence evidence.

Correlation units are not the same as protected components

Correlation-unit membership is only one way that two frames can be scientifically unsafe to separate. Define

$$R_{\text{prot}} = R_{\text{corr}} \cup R_{\text{dup}} \cup R_{\text{event}} \cup R_{\text{replica}} \cup R_{\text{realization}},$$

representing correlation-unit membership, exact-geometry duplication, protected-event membership, condition-scoped replica lineage, and condition-scoped structural-realization lineage. The actual no-split relation is the transitive closure

$$\sim_{\text{prot}} = \text{TC}(R_{\text{prot}}).$$

Thus, if frame a shares a correlation unit with b , and b is an exact duplicate of c , then a, b, c belong to one indivisible protected component even if no direct a - c relation was recorded.

Two distinct objects. The **correlation unit** is the autocorrelation-aware temporal/statistical block and is the unit balanced by the coverage measure. The larger **protected component** is the connected component of all accepted no-split relations and is the object that cannot be divided across incompatible evidence roles or across the P_{train}/M_3 boundary.

From protected units to the exact P_{train} population

Outer evidence roles are assigned before target selection

The neutral statistical substrate assigns complete units to protected outer roles rather than splitting individual frames. These roles include development, outer monitor, uncertainty calibration, locked interpolation test, purge, and exclusion.

The current construction deterministically places monitor/calibration/locked anchors where available conditions support them, purges neighboring same-run units according to the role policy, and leaves the remaining admissible units as **DEVELOPMENT** evidence. Thus outer evidence is separated before target-size membership is selected.

Only exact eligible DEVELOPMENT frames with canonical training labels enter

$$U_{\text{size}}.$$

The target-training selector never draws from outer monitor, calibration, locked-test, purged, or excluded evidence.

Exact protected split into P_{train} and M_3

Project $\sim_{\text{prot}}$ onto U_{size} and compute its connected components. Those components are indivisible allocation units.

The evaluation reserve must contain *exactly* M_3 frames. If protected component j contains w_j frames, allocation is an exact 0/1 subset-sum problem. With reachable cardinalities R_j ,

$$R_0 = \{0\}, \quad R_j = R_{j-1} \cup \{r + w_j : r \in R_{j-1}, r + w_j \leq M_3\}.$$

Components are processed in a deterministic order balancing component size and condition identity; the accepted first-predecessor rule fixes one exact membership when several solutions exist. If exact M_3 cardinality is impossible without splitting a protected component, preparation fails rather than weakening the protection rule.

The selected components form M_3 ; their complement forms

$$P_{\text{train}} = U_{\text{size}} \setminus M_3.$$

This is the frozen target-order domain. From this point onward, selector-specific fitting, scaling, neighborhoods, coverage, and ordering are constructed on exact P_{train} only.

One nested training-set ladder

Let

$$P_{\text{train}} = \{x_1, \dots, x_n\}.$$

`mdstats` constructs one deterministic permutation

$$\pi_{\text{train}} = (x_{(1)}, x_{(2)}, \dots, x_{(n)}),$$

and

$$T_N = \pi_{\text{train}}[:N].$$

Hence

$$N_a < N_b \implies T_{N_a} \subset T_{N_b}.$$

The learning-curve experiment is nested: increasing N retains all earlier configurations and adds new ones chosen to improve physical representation. No independent per- N selector is run.

Multi-view construction of the physical feature space

For atomistic data there is no unique scalar notion of “distance between configurations.” Two structures can be similar in pair distances but different in angular order; they can have similar geometry but very different forces, pressure, or strain. A single concatenated descriptor can also mix quantities with unrelated physical meanings and scales.

`mdstats` therefore uses a **multi-view representation**. For each required feature family m ,

$$\mathbf{f}_m(x) = (f_{m1}(x), \dots, f_{md_m}(x)).$$

Physical feature families used to represent the target-training population

Physical view	Representative information
Local structure	Pair distance, radial environment, coordination, connectivity, chemical environment, local density, angular environment, orientational order.
Pair geometry	Minimum pair distance, mean and maximum nearest-neighbor distance, and coordination statistics for applicable pair rules.
Target response	Force statistics, energy per atom, temperature, hydrostatic/deviatoric strain, pressure, and stress-deviator magnitude when defined.
Profile-specific evidence	Accepted continuous selection features and discrete environment classes supplied by an active material/profile provider.
Foundation-model weakness	Energy and force residuals only when target-size training itself uses an authenticated frozen foundation model.

Each family is interpreted independently; one well-covered physical view cannot compensate for a poorly covered one.

Robust scaling inside each family

For feature coordinate j , use the interquartile range when non-degenerate,

$$s_j = Q_j(0.75) - Q_j(0.25),$$

with broader-quantile and variance-based fallbacks for nearly constant coordinates. The normalized family distance is

$$d_m(a, b) = \sqrt{\frac{1}{d_m} \sum_{j=1}^{d_m} \left(\frac{f_{mj}(a) - f_{mj}(b)}{s_j} \right)^2}.$$

The factor $1/d_m$ prevents a family from gaining influence merely by having more coordinates.

Coverage as a local covering problem

For a witness w in family m , define an adaptive local radius $r_m(w)$: the smallest distance containing approximately

$$\beta = \frac{1}{128}$$

of the remaining correlation-balanced reference mass. Dense regions therefore receive smaller neighborhoods and sparse regions larger ones.

Candidate c represents witness w when

$$A_m(w, c) = 1 \iff d_m(w, c) \leq r_m(w),$$

up to the prescribed numerical tolerance. For selected set S ,

$$n_m(w; S) = \sum_{c \in S} A_m(w, c),$$

and the witness is covered when $n_m(w; S) > 0$.

Correlation-balanced reference mass

Let $g(w)$ be the P1 correlation-unit identity of witness w , let G_m be the represented units in family m , and let $n_{m,g}$ be the witness count from unit g . Then

$$\omega_m(w) = \frac{1}{|G_m| n_{m,g(w)}}.$$

Every represented correlation unit therefore receives exactly the same total mass in that feature family, regardless of how many stored frames it contains; frames inside the unit divide that mass.

The covered mass is

$$C_m(S) = \sum_{w \in W_m} \omega_m(w) \mathbf{1}[n_m(w; S) > 0],$$

with hard baseline

$$C_m(S) \geq 0.95 \quad \text{for every required feature family}.$$

Thus 95% coverage means that at least 95% of the *correlation-balanced reference mass* in each physical view has a selected representative nearby. It does not mean 95% model accuracy, 95% confidence, or 95% of raw trajectory frames.

Protecting physically important tails and discrete support

For extent-bearing channel j ,

$$L_j = Q_j(0.01), \quad U_j = Q_j(0.99),$$

and the selected set must satisfy

$$\min_{x \in S} f_j(x) \leq L_j, \quad \max_{x \in S} f_j(x) \geq U_j.$$

Discrete **hard support obligations** additionally require applicable thermodynamic conditions, structural-event classes, active profile environment classes, both extent sides, and each represented current P1 correlation unit. These requirements cannot be traded away for better average coverage elsewhere.

How coverage is achieved algorithmically

For candidate c , newly covered mass in family m is

$$G_m(c) = \sum_{\substack{w: A_m(w,c)=1 \\ n_m(w;S)=0}} \omega_m(w).$$

A large $G_m(c)$ means that c fills a genuine hole in the current representation.

Stage 1: satisfy obligations and fill the weakest-covered view

While any hard obligation is unsatisfied or any required family has $C_m < 0.95$, candidates are prioritized lexicographically to:

1. satisfy as many currently missing hard obligations as possible;
2. maximize new coverage in the least-covered required feature family;
3. maximize new coverage summed over all families;
4. favor correlation units currently underrepresented in the prefix;

5. prefer broader representative utility and sparse diversity; and
6. use stable frame identity only as the final deterministic tie-breaker.

The algorithm is therefore **bottleneck driven**: a well-covered family cannot compensate for a poorly covered family.

Stage 2: densify the covered manifold

After all hard requirements and family thresholds pass, selection continues using diminishing-return representative utility

$$R(c) = \sum_m \sum_{w: A_m(w,c)=1} \frac{\omega_m(w)}{n_m(w; S) + 1}.$$

The first representative of a region is most valuable; the second is still useful; additional representatives contribute progressively less. Selection therefore changes smoothly from filling missing regions to increasing sampling density over the already covered physical manifold.

Configured sizes, qualification, and physical interpretation

Requested sizes $N_1 < N_2 < \dots < N_K$ are prefixes of the same master order. Limited deterministic repair may alter only the newly added shell near a configured boundary; earlier configured prefixes remain immutable.

Each prefix is independently qualified. It may enter target-size model training only if the exact prefix exists, required labels are usable, every required feature family passes coverage, all required lower/upper extents are represented, and all hard support obligations reach their minima.

The method therefore constructs an increasingly fine **cover of the accessible training population** while preserving leakage control upstream. A useful configuration supplies mandatory support, covers a previously unrepresented region, or increases representation density in a sparsely represented region.

number of configurations $\neq$ coverage of physical variation $\neq$ MLFF prediction accuracy

Coverage is a property of training-set membership. Prediction accuracy is measured later on independent evaluation evidence. The resulting experiment is:

As the number of training configurations increases, while protected dependence is respected and multi-view physical coverage remains balanced, how much additional predictive accuracy does the MLFF gain?

Authority and external scientific context

This summary is explanatory and non-authoritative. The upstream statistical/data-preparation semantics and target-order semantics are owned principally by:

- docs/methods/mlff_scientific_method.md and docs/methods/mlff_numerical_algorithmic_method.md for correlation, protected dependence, evidence roles, and the exact $U_{\text{size}} \rightarrow P_{\text{train}} + M_3$ split;
- docs/methods/mlff_target_training_order_scientific_method.md and docs/methods/mlff_target_training_order_numerical_algorithmic_method.md for multi-view coverage, obligations, ordering, repair, and qualification; and

- `docs/specs/sampling/shared_sampling_primitives_spec.md` and `docs/specs/training_data/mlff_data5_partition_roles_spec.md` for the implemented sampling/partition contracts summarized here.

External literature provides scientific context but does **not** define the project-specific `mdstats` thresholds, feature families, split rules, or ordering semantics.

1. H. Flyvbjerg and H. G. Petersen, “Error Estimates on Averages of Correlated Data,” *Journal of Chemical Physics* **91**, 461–466 (1989). doi:10.1063/1.457480.
2. C. J. Geyer, “Practical Markov Chain Monte Carlo,” *Statistical Science* **7**, 473–483 (1992). doi:10.1214/ss/1177011137.
3. J. Behler and M. Parrinello, “Generalized Neural-Network Representation of High-Dimensional Potential-Energy Surfaces,” *Physical Review Letters* **98**, 146401 (2007). doi:10.1103/PhysRevLett.98.146401.
4. A. P. Bartok, R. Kondor, and G. Csanyi, “On representing chemical environments,” *Physical Review B* **87**, 184115 (2013). doi:10.1103/PhysRevB.87.184115.
5. V. L. Deringer, M. A. Caro, and G. Csanyi, “Machine Learning Interatomic Potentials as Emerging Tools for Materials Science,” *Advanced Materials* **31**, 1902765 (2019). doi:10.1002/adma.201902765.
6. E. V. Podryabinkin and A. V. Shapeev, “Active learning of linearly parametrized interatomic potentials,” *Computational Materials Science* **140**, 171–180 (2017). doi:10.1016/j.commatsci.2017.08.031.

Method summary aligned with the current accepted `mdstats` correlation/statistical-preparation and target-training-order authority chain. Implementation-only storage, scheduler, sparse-layout, and acceleration details are omitted where they do not change scientific membership semantics.